



Hydroxyurea inhibits parvovirus B19 replication in erythroid progenitor cells



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ABSTRACT

Parvovirus B19 (B19V) infection is restricted to erythroid progenitor cells (EPCs) of the human bone marrow, leading to transient arrest of erythropoiesis and severe complications mainly in subjects with underlying hematological disorders or with immune system deficits. Currently, there are no specific antiviral drugs for B19V treatment, but identification of compounds inhibiting B19V replication can be pursued by a drug repositioning strategy. In this frame, the present study investigates the activity of hydroxyurea (HU), the only disease-modifying therapy approved for sickle cell disease (SCD), towards B19V replication in the two relevant cellular systems, the UT7/EpoS1 cell line and EPCs. Results demonstrate that HU inhibits B19V replication with EC₅₀ values of 96.2 μ M and 147.1 μ M in UT7/EpoS1 and EPCs, respectively, providing experimental evidence of the antiviral activity of HU towards B19V replication, and confirming the efficacy of a drug discovery process by drug repositioning strategy. The antiviral activity occurs *in vitro* at concentrations lower than those affecting cellular DNA replication and viability, and at levels measured in plasma samples of SCD patients undergoing HU therapy. HU might determine a dual beneficial effect on SCD patients, not only for the treatment of the disease but also towards a virus responsible for severe complications.

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1. Introduction

Parvovirus B19 (B19V) is a human virus in the Parvoviridae family with an ample pathogenic potential [1–4]. Common clinical manifestations are erythema infectiosum in children, arthropathies mainly in adults, however the virus has been associated to many other different pathologies and inflammatory processes affecting diverse tissues and organs. A main characteristic of B19V is its selective tropism for erythroid progenitor cells (EPCs) in the bone marrow, or the equivalent during fetal development. As a consequence, B19V infection can result in chronic pure red blood cell aplasia mostly in individuals with compromised immune system or transient aplastic crisis (TAC) in patients under conditions of hematopoietic stress in which there is increased red cell destruction worsening anemia [5,6]. This complication is particularly common in young children with sickle cell disease (SCD), and less frequent, but present, in adults [7–9]. In these subjects, red blood cell (RBC) transfusions are adequate to treat a transient aplastic episode reducing the risk of circulatory collapse from severe anemia, or other B19V-related events (marrow necrosis, acute splenic

sequestration, acute chest syndrome, fat embolism) [10–12]. RBC transfusions are also indicated to overcome chronic anemia and, in case of fetal infection, intrauterine transfusions are a therapeutic option in the treatment of fetal anemia and non-immune hydrops associated to B19V [13]. Chronic anemia due to persistent infection is responsive to intravenous immunoglobulin (IVIG) administrations, but high dosage and repeated courses may be necessary, and symptoms can recur when IVIG treatment is interrupted [14,15]. Currently, there are no recognized antiviral drugs as an alternative to the supportive treatments however in certain B19V-infected patients a specific therapy would provide a substantial benefit preventing bone marrow failure or inflammatory aspects of the infection.

The absence of the complete characterization of B19V proteins and related cellular machinery interactions makes difficult a rational drug design strategy. The NS protein, with its replicative activities and effects on cellular environment, and the phospholipase domain in the VP1u region may constitute relevant therapeutic targets [3,4], however these proteins have not been characterized to the availability of crystallographic structures, and entries in UniprotKB database only refer to predicted domains [16], thus preventing accurate target-based molecular modeling. Also, the interactions of viral proteins with cellular counterparts

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are only partially described and understood [3,4]. Thus, efforts in the search for compounds inhibiting B19V replication can be aptly directed towards a drug repositioning approach. anti-B19V drug discovery can take advantage of the availability of two cellular systems, the myeloblastoid cell line UT7/EpoS1 and the human primary EPCs able to support B19V replication [17], thus suitable to test antiviral drugs. On the other hand, animal *in vivo* models, although possible, are not easily amenable to experimental use for this purpose [18,19].

In this frame, the anti-B19V activity of cidofovir has been demonstrated, widening the antiviral spectrum of the acyclic nucleoside phosphonate [20,21]. The drug exerts a potent antiviral effect in infected myeloblastoid cell line UT7/EpoS1 and a relevant inhibitory activity in primary human EPCs when extendedly exposed to high doses [22,23], thus limiting its potential use *in vivo*.

This evidence encourages the search for other anti-B19V candidates and, herein, the repositioning opportunity of hydroxyurea (HU) in the treatment of B19V infection was evaluated. HU is an S-phase inhibitor of DNA synthesis targeting cellular ribonucleotide reductase (RR) enzyme [24], thus leading to the inhibition of cell proliferation, including the hematopoietic cell compartment that is involved in the pathogenesis of B19V infection. The potential use of HU as antiviral agent has been previously explored both *in vitro* and *in vivo* in combined therapy with deoxynucleoside analogs with the assumption that HU depletes the intracellular deoxyribonucleotide pools required for viral replication, enhancing deoxynucleoside analogs incorporation [25–28]. Finally, the drug is the only disease-modifying therapy approved by FDA for SCD in adults [29,30], and its “off-label” use is increasing in the pediatric SCD population [31–35], in which B19V is cause of severe complications.

Given all these observations, the present study investigates the possible inhibitory effects of HU against B19V replication in the two relevant model cellular systems, UT7/EpoS1 cells and in EPCs, by means of quantitative evaluation of viral nucleic acids synthesized within a time course of infection. In addition, a complete characterization of the cellular effects of HU in the selected experimental systems was carried out including viability, proliferation, and cytotoxicity, together with the immunophenotypic characterization of erythroid-associated markers and cell cycle analysis by flow cytometry in the EPCs.

2. Materials and methods

2.1. Drug

Hydroxyurea (HU) was purchased from Sigma-Aldrich (St. Louis, MO, USA) at a purity degree >98%, solubilized in water and filter sterilized. Aliquots were stored at -20°C and used in the experiments in the range 0.1–50 mM.

2.2. Cells

UT7/EpoS1 cells were cultured in Iscove modified Dulbecco medium (IMDM) (Sigma-Aldrich, St. Louis, MO, USA), containing 10% fetal bovine serum (Carlo Erba Reagents, Milan, Italy) and 2 U/mL rhu erythropoietin (NeoRecormon, Roche S.p.A, Monza, Italy), at 37°C and 5% CO_2 . Cells were used for experiments 24 h after a fresh medium exchange, when at a density of $3\text{--}5 \times 10^5$ cells/mL.

For the generation of EPCs from peripheral blood mononuclear cells (PBMC), a previously established protocol was adopted [17]. PBMC were obtained from leukocyte-enriched buffy coats of anonymous blood donor, available for institutional research purposes from the Immunohematology and Transfusion Service

(S. Orsola-Malpighi University Hospital, Bologna, Italy, authorization 0070755/1980/2014, issued by Head of Service). Following isolation by Ficoll-Paque centrifugation (GE Healthcare Bio-Science AB, Uppsala, Sweden), PBMC were cultured in IMDM supplemented with 20% serum substitute BIT 9500 (StemCell Technologies, Vancouver, Canada), and enriched with 900 ng/mL ferrous sulphate, 90 ng/mL ferric nitrate, 1 μM hydrocortisone, 3 U/mL rhu erythropoietin, 5 ng/mL rhu IL-3 and 20 ng/mL rhu stem cell factor (SCF) (Life Technologies, Carlsband, CA, USA). Cells were maintained at 37°C in 5% CO_2 and kept at 10^6 cells/mL until experiments.

2.3. Cell proliferation and cell viability assays

Cell proliferation measurements were assessed in UT7/EpoS1 cells and EPCs, at 8 ± 2 days from isolation, by evaluating the 5-bromo-2'-deoxyuridine (BrdU) incorporation into the newly synthesized genomic DNA in a 48 h-period (Cell proliferation ELISA BrdU Assay, Roche Diagnostics, Indianapolis, IN, USA). For analysis, cells at a density of 5×10^4 cells/100 μL were seeded in 96-well plate and grown in regular medium or supplemented with different concentrations of HU (0.1–50 mM). At 24 h, 10 μM of BrdU were added to cells, the cells were harvested at 48 h and the amount of labeled DNA in the cell population was measured according to manufacturer's instructions. Data were expressed as mean percentage values of treated cells relative to untreated controls.

The assessment of cellular viability and metabolic function was carried out through the measurement of mitochondrial dehydrogenase enzyme activity by the WST-8 based assay (Cell Counting Kit-8, CCK-8, Dojindo Molecular Technologies, Rockville, MD, USA). For these experiments, UT7/EpoS1 cells and EPCs were seeded at the density of 5×10^4 cells/100 μL in a 96-well plate, and cultured for 48 h with the different concentrations of HU (0.1–50 mM). After addition of the CCK-8 solution, the amount of the formazan dye generated by dehydrogenases in living cells was measured according to manufacturer's instructions. Data were expressed as mean percentage values of treated cells relative to untreated controls.

The assessment of potential cytotoxic properties of HU on the investigated cellular systems was carried out by measuring the lactate dehydrogenase enzyme (LDH) released in the cell culture medium through damaged plasma membranes (Cytotoxicity LDH Assay kit-WST, Dojindo Molecular Technologies, Rockville, MD, USA). For the assay, cells were cultured at a density of 5×10^4 cells/100 μL in a 96-well plate with different concentrations of HU (0.1–50 mM), and, following 48 h of incubation, the cell-free supernatants (100 μL) were recovered from each well, and the LDH activity was determined following manufacturer's instructions. HU cytotoxicity was expressed as percentage values relative to the 100% lysis controls, included in the test, and untreated controls.

2.4. B19V infection

UT7/EpoS1 cells and EPCs, at 8 ± 2 days from isolation, were infected with B19V at a multiplicity of infection (moi) of 10^4 viral genomes/cell. The inoculum virus consisted in a B19V viremic serum sample, containing 10^{12} B19V (genotype 1) genome copies (geq)/mL, as determined by qPCR assay [36], available for research purposes according to the Italian privacy law. The viremic serum resulted negative to B19V specific IgM and IgG antibodies by a commercial chemiluminescence immunoassay (Liaison XL, Diasorin S.p.A, Italy) and to other viruses (HIV, HBV, HCV, HSV, VZV, EBV, CMV, HHV8, AdV, BKV) by nucleic acid testing (NAT) assays.

Cells at 10^7 /mL were incubated with B19V for 2 h at 37°C, then washed with PBS, and cultured at 10^6 cells/mL in the respective fresh medium supplemented with or without increasing concentrations of HU (0.1–50 mM).

2.5. Analysis of B19V nucleic acids by qPCR assays

Amounts of cell cultures, corresponding to 1.5×10^5 cells, were collected at 2 hpi and 48 hpi (hours post-infection) for nucleic acids purification by using the EZ1 viral nucleic acid kit on a EZ1 platform (Qiagen Hilden, Germany), following the manufacturer's instructions. The quantitative evaluation of viral nucleic acids was carried out by qPCR assays in a Rotor-Q system (Qiagen, Hilden, Germany). For the analysis of B19V DNA, aliquots of the eluted nucleic acids (corresponding to 1.5×10^4 cells) were directly amplified and detected through a Sybr Green technology (Maxima SYBR Green qPCR Master Mix, Thermo Scientific, Waltham, MA, USA), while samples for B19V RNA analysis were treated with a Turbo DNasefree reagent (Ambion, Thermo Scientific, Waltham, MA, USA) before amplification in a qRT-PCR assay (Express One-step SYBR GreenER Kit, Invitrogen Corporation, Carlsbad, CA, USA). Standard cycling programs according to manufacturer's protocols were used and followed by a melting curve analysis to define the melting temperature (T_m) of products as a confirmation of qPCR specificity. Primers for the amplification of viral targets were the pair R2210–R2355, located in the central exon of B19V genome, suitable for amplification of both DNA and the whole set of viral transcripts [37]. Absolute quantitation of viral targets was obtained using external calibration curves, and expressed as DNA or RNA target copies/ 1.5×10^4 cells. In addition, a target sequence in the region of genomic DNA coding for 5.8S rRNA (rDNA) was amplified by using the primer pair 5.8S_S1 (CTCTTAGCGGTGGATCACTC)/5.8S_A1 (GTGCGTTTCAAGTGTGATG), as internal control of nucleic acid purification and amplification processes.

2.6. Flow cytometric analysis

In vitro cultured EPCs, at 8 ± 2 days from isolation, were characterized using flow cytometry (FACSCalibur, Becton Dickinson, Franklin Lakes, NJ, USA) before and following a 48-h time course of B19V infection and/or treatment with selected HU concentrations. Samples were gated on the forward light scatter (FSC) versus side light scatter (SSC) plot to exclude cellular debris and were analyzed using the Cell Quest Pro software (Becton Dickinson, Franklin Lakes, NJ, USA). For cell surface antigen analysis, EPCs were stained with antibodies specific for erythroid lineage differentiation markers (CD36-PE, CD71-PE, CD235a-FITC, BD Biosciences, San Jose, CA, USA), following a previously established protocol [17]. Data are presented as the mean values $\pm$ standard error of two separate experiments. For cell cycle analysis, after a fixation step in cold 70% ethanol, EPCs were incubated in a propidium iodide (PI) staining solution containing 0.1% Triton X-100 (Sigma-Aldrich, St. Louis, MO, USA), 50 μ g/mL PI and 100 μ g/mL RNase in PBS, for 30 min at 37°C. Pulse-width vs pulse-area plot was used to exclude cell doublet then the gate was applied on PI histogram plot; the percentage of cells in each cell cycle phase was quantitatively evaluated by setting markers within the analysis program. Data are reported as mean percentage values in two independent experiments.

2.7. Statistical analysis

Results are presented as the mean values $\pm$ standard error of at least two separate experiments, each performed at least in duplicate, except when indicated. Percentage values of samples at the different experimental conditions were determined as relative to untreated sample controls. Fifty-percent cytotoxic concentration

(CC₅₀) values were determined using the program GraphPad Prism version 5.00 for Windows (GraphPad Software, San Diego, CA, USA) as the inflection points of the four-parameter dose-response curves obtained by plotting the percentage values of the inhibition of BrdU incorporation or formazan dye reduction vs. the logarithm of HU concentration. Likewise, 50% effective concentration (EC₅₀) values were obtained by plotting the percentage values of the inhibition of B19V replication vs. the logarithm of HU concentration.

One-way analysis of variance (ANOVA) followed by Dunnett's Multiple comparison test was used to compare data among the different experimental conditions; statistically significant differences were determined at $p < 0.05$.

3. Results and discussion

3.1. Cytostatic and cytotoxic effects of HU in UT7/EpoS1 and EPCs

An accurate prediction of the effects of HU on UT7/EpoS1 cell line and in EPCs generated *in vitro* from peripheral blood mononuclear cells was carried out including multiple toxicity end-point determinations. The cytostatic effect was assessed in a 48 h-period by the measurement of DNA synthesis by BrdU incorporation and mitochondrial dehydrogenase enzyme activity by a WST-8 based method, while the cytotoxicity was evaluated by the determination of the loss of membrane integrity by LDH release.

In both experimental systems cellular DNA replication was affected in a dose-dependent manner with HU concentrations leading to a 50% reduction of DNA synthesis at 706.9 μ M (95% Confidence Interval: 579.5–880.4 μ M; R^2 0.95) and 494.0 μ M (95% Confidence Interval: 422.2–571.5 μ M; R^2 0.95) in UT7/EpoS1 and EPCs, respectively (Fig. 1A). The observed cytostatic effect of HU was confirmed by measurement of formazan dye reduction using the WST-8 based method, in both systems. The assay demonstrated a dose-dependent reduction of the overall metabolic activity with values comparable with those obtained by measuring the total amount of BrdU labeled DNA in the populations. The 50% HU-induced reduction of viability occurred at 581.9 μ M (95% Confidence Interval: 426.5–812.3 μ M; R^2 0.95) for UT7/EpoS1, and 584.8 μ M (95% Confidence Interval: 478.5–722.3 μ M; R^2 0.95) for EPCs (Fig. 1B).

The cytotoxicity LDH assay was performed on the cell-free supernatant of investigated cellular populations treated for 48 h with HU at the different concentrations; at all tested concentrations, UT7/EpoS1 did not display loss of membrane integrity indicating that the observed reduction of cell viability could be entirely ascribed to the cytostatic effect of the drug. In contrast, measurements of LDH activity revealed that EPCs membranes lost in integrity for HU concentrations ≥ 500 μ M but never reaching a 50% of cell damage: cytotoxicity values ranged from $25.6 \pm 1.5\%$ at 500 μ M to $40.4 \pm 4.6\%$ at 50 mM. Even in this cellular system, consisting in a heterogeneous population of primary cells, HU exerted mainly cytostatic properties rather than cytotoxic effects, in agreement with its inhibitory activity on RR enzyme and consequently on cell proliferation [24].

3.2. Antiviral activity of HU in UT7/EpoS1 and EPCs

B19V nucleic acids amounts were quantitatively evaluated within infected cells by means of qPCR assays, at 2 hpi and 48 hpi, and for the different HU concentrations (0–50 mM). In both cellular systems, a progressive decrease of B19V DNA amounts in dependence of HU was observed, with DNA amounts obtained from cells treated with ≥ 1 mM of HU lower than input DNA (2 hpi), indicating absence or negligible replicative activity of B19V

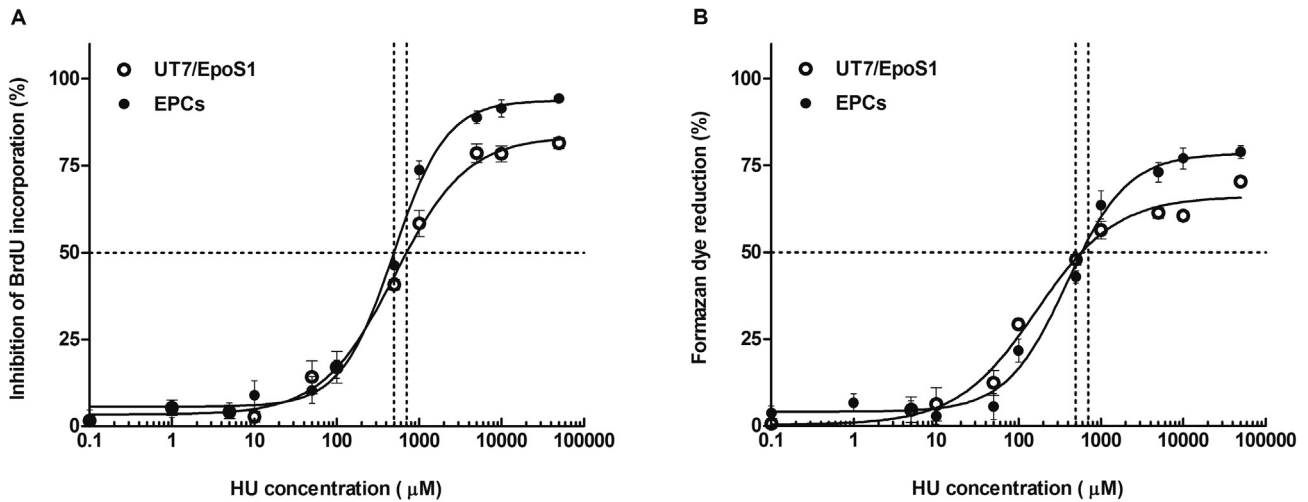


Fig. 1. Effects of HU on viability in UT7/EpoS1 and EPCs. (A) Concentration-dependent effects of HU on cellular DNA synthesis (BrdU incorporation assay). Fifty-percent inhibition of cellular DNA replication in response to HU treatment is at 706.9 μM (95% Confidence Interval: 579.5–880.4 μM; R^2 0.95) for UT7/EpoS1, and 494.0 μM (95% Confidence Interval: 422.2–571.5 μM; R^2 0.95) for EPCs. Data were obtained by three separate experiments, each performed in triplicate. (B) Concentration-dependent effects of HU on cell viability (CCK-8 assay). Fifty-percent HU-induced reduction of viability occurs at 581.9 μM (95% Confidence Interval: 426.5–812.3 μM; R^2 0.95) for UT7/EpoS1, and 584.8 μM (95% Confidence Interval: 478.5–722.3 μM; R^2 0.95) for EPCs. Data were obtained by two separate experiments, each performed in triplicate.

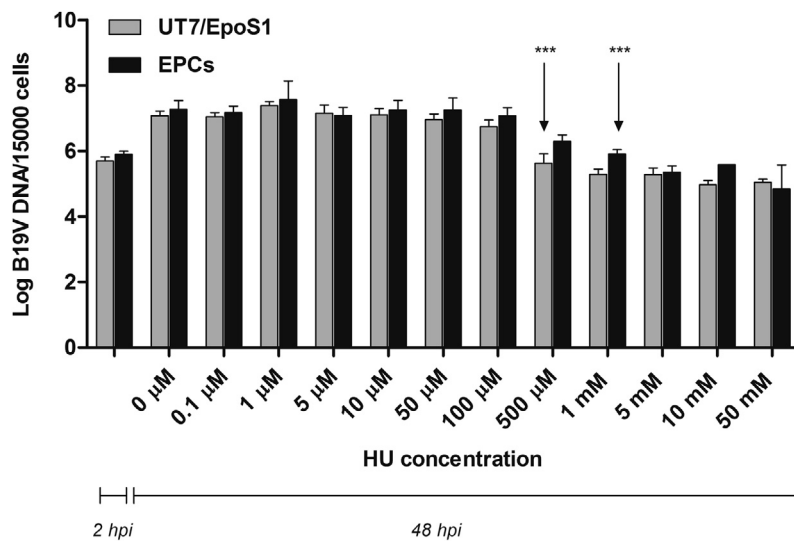


Fig. 2. B19V DNA in UT7/EpoS1 cells and EPCs. The amounts of B19V DNA were determined by qPCR, at 2 and 48 hpi, at the different HU concentrations, and expressed as Log B19V DNA target copies/15,000 cells. Histograms are mean values $\pm$ standard error of means obtained from 5 independent experiments. Analysis of variances followed by Dunnett's Multiple comparison test indicate a significant reduction in B19V replication compared to untreated cells for UT7/EpoS1 at HU $\geq$ 500 μM and for EPCs at HU $\geq$ 1 mM ($^{***}p < 0.0001$).

(Fig. 2). By expressing the dose dependent relationship between HU and B19V DNA amounts as percentage inhibition of replication relative to the untreated controls (0 mM HU), non-linear regression curves were generated yielding to EC_{50} values of 96.2 μM (95% Confidence Interval: 81.5–118.1 μM; R^2 0.94) and 147.1 μM (95% Confidence Interval: 121.4–190.4 μM; R^2 0.95) for UT7/EpoS1 and EPCs, respectively (Fig. 3). Quantitative results obtained by RT-qPCR assay demonstrated that, in both cellular systems at all tested HU concentrations, viral transcription occurred at levels higher than those measured at 2 hpi (Fig. 4). However, B19V RNA amounts progressively reduced comparing to untreated cells at the highest HU concentrations indicating an association between levels of B19V DNA template and viral transcription. In particular, in UT7/EpoS1 cells RNA amounts statistically decreased for HU concentrations $\geq$ 1 mM while in EPCs for concentrations $\geq$ 10 mM ($p < 0.0001$).

The effective values inhibiting viral replication, consistent with those obtained for other human pathogenic viruses [26], are lower than those interfering with cell proliferation indicating that HU acts as an antiviral agent towards B19V replication in both systems at similar levels. It is conceivable that, due to the highly dynamic nature of viral replication, inhibition of HU might preferentially disturb viral rather than cellular synthesis. The acyclic nucleoside phosphonate cidofovir, the only other antiviral drug with demonstrated activity against B19V replication *in vitro*, proved to be effective in UT7/EpoS1 but in EPCs displayed activity only at high concentration and upon extended exposure [22,23].

Importantly, SCD patients undergoing HU therapy experience peak plasma concentrations of 276–709 μM [38], indicating that HU levels sufficient to reduce B19V replication are achievable *in vivo*. At the indicated HU plasma concentrations the antiviral effect of the drug increases considerably without significant

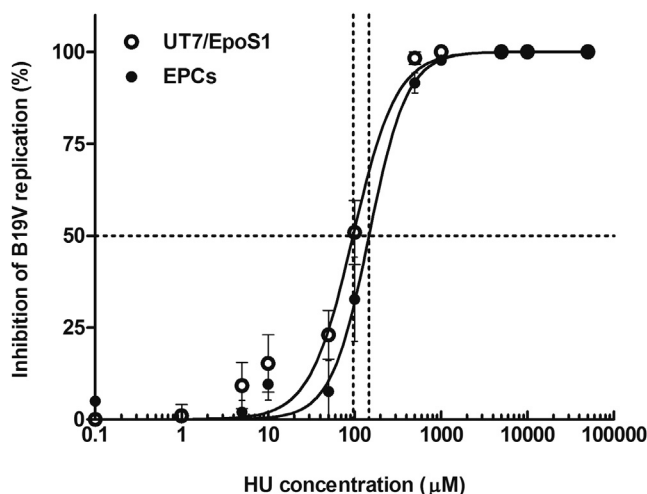


Fig. 3. B19V replication in UT7/EpoS1 cells and EPCs in response to the different HU concentrations. Plots are the non-linear regression curves obtained by expressing B19V DNA amounts as percentage of inhibition relative to the untreated controls. For UT7/EpoS1 cells: EC_{50} value is 96.2 μ M (95% Confidence Interval: 81.5–118.1 μ M; R^2 0.94); for EPCs EC_{50} value is 147.1 μ M (95% Confidence Interval: 121.4–190.4 μ M; R^2 0.95). Results are obtained from 5 independent experiments.

toxicity. The estimated percentages of inhibition towards B19V replication range from 77.7 to 95.8%, interpolating HU concentrations on the dose–response curve for EPCs. Concerning the effects of HU on the overall cellular health, at the indicated range of concentrations, HU still displayed a cytostatic (33.4–60.6%) rather than cytotoxic effect (<30%).

3.3. Effects of HU on EPCs population

In order to gain insight into the effects of HU on a proliferating and differentiating cellular population, the expression of the markers of intermediate erythroid differentiation process, CD36, CD71 and CD235a, was assessed by flow cytometric analysis in EPCs when incubated for 48 h with selected drug concentrations, corre-

sponding to the 50% cytostatic concentration, as determined by BrdU incorporation, and the EC_{50} for B19V replication (494.0 μ M and 147.1 μ M, respectively). The same set of experiments was carried out on B19V infected cells to evaluate a possible combined effect. Flow cytometry dot plots showing the distribution of cell populations with respect to forward (FSC) and side scatter (SSC) parameters, at the different experimental conditions, are reported in Fig. 5.

EPCs were analyzed after 8 days from isolation for CD36, CD71 and CD235a markers, and after 2 days of treatment with HU or without as control (Fig. 6A). During this time an increase in positive cells expressing the specific markers was observed in the cell population grown in the absence of the drug, related to the differentiation and maturation of cell population along the erythroid lineage. Indeed, a statistically significant expansion of CD36 and CD71 positive cell population was measured in this time-lapse ($p < 0.001$). By contrast, EPCs cultured in medium supplemented with HU displayed a dose-dependent reduction of the fraction of positive cells compared to the untreated control and a statistically significant difference was observed for CD71 and CD235a markers in treated cells with 494.0 μ M HU ($p < 0.05$). These findings corroborate the overall data on EPCs metabolism and viability obtained with the BrdU incorporation assay and CCK-8 method, confirming the cytostatic effect of HU at the indicated concentrations and its inhibitory activity on hematopoietic cell proliferation.

EPCs were also phenotypically characterized following a time course of B19V infection, in the absence and in the presence of HU at 494.0 μ M and 147.1 μ M (Fig. 6B). The B19V infected culture grown in regular medium displayed percentage values of positive cells to CD36, CD71 and CD235a markers not different from those determined two days in advance, indicating that viral replication completely inhibited the expansion of EPCs population. This viral-induced arrest of erythropoiesis is responsible *in vivo* of the transient or persistent erythroid aplasia. In addition, HU treatment following B19V infection produced a further reduction of the fraction of EPCs positive cells. As a consequence of the relatively high biological variability of the differentiation progress in the cell population, a statistically significant difference was evidenced for the expression of CD71 cell-surface antigen in EPCs treated with 494.0 μ M HU compared to untreated but B19V infected cells

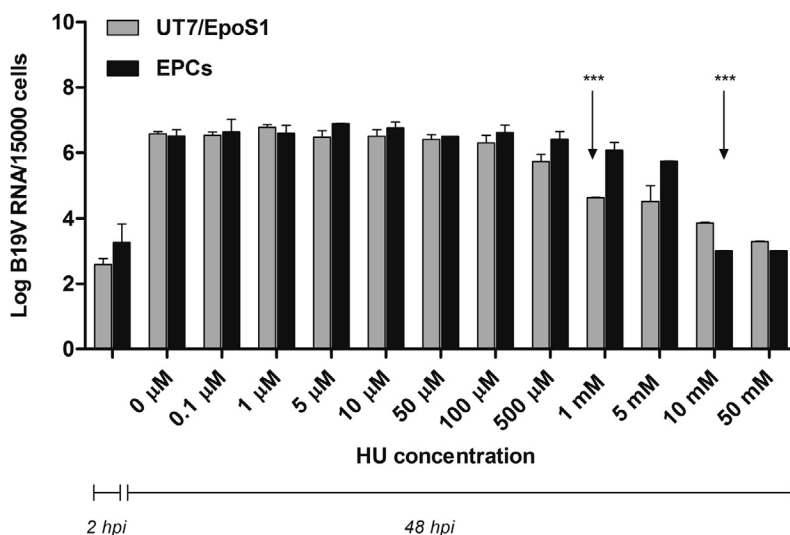


Fig. 4. B19V RNA in UT7/EpoS1 cells and EPCs. The amounts of B19V total RNA were determined by qRT-PCR, at 2 and 48 hpi, at the different HU concentrations, and expressed as Log B19V RNA target copies/15,000 cells. Histograms are mean values $\pm$ standard error of means obtained from 2 independent experiments. Analysis of variances followed by Dunnett's Multiple comparison test indicate a significant reduction in transcriptional activity compared to untreated cells for UT7/EpoS1 at HU $\geq$ 1 mM and for EPCs at HU $\geq$ 10 mM ($***p < 0.0001$).

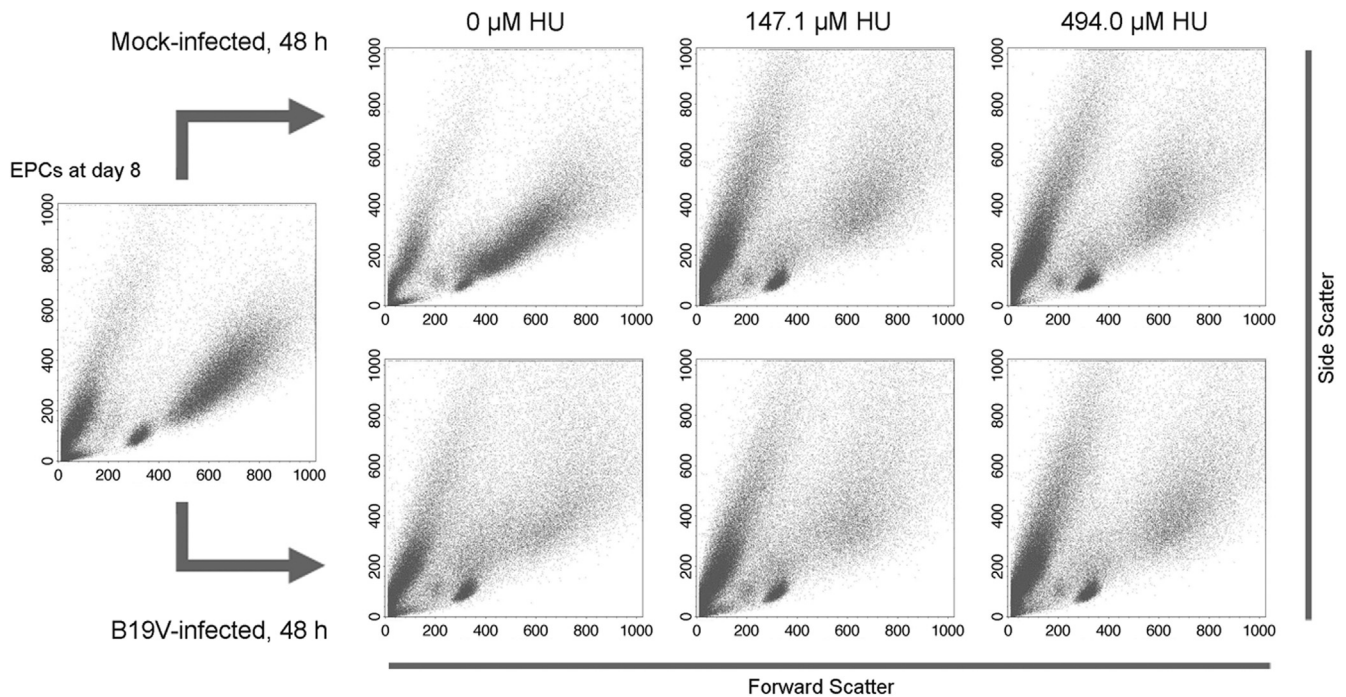


Fig. 5. Changes in populations of EPCs on two-dimensional dot plots upon HU treatment and B19V infection. Untreated and mock-infected EPCs displayed different physical properties from all other experimental conditions indicating that both HU and viral replication disrupt the maturation process along the erythroid lineage as visible from variations in the cellular morphology of EPCs population. Dot plots represent forward scatter vs side scatter and shown representative data from two independent experiments performed on three different leukocyte-enriched buffy coats.

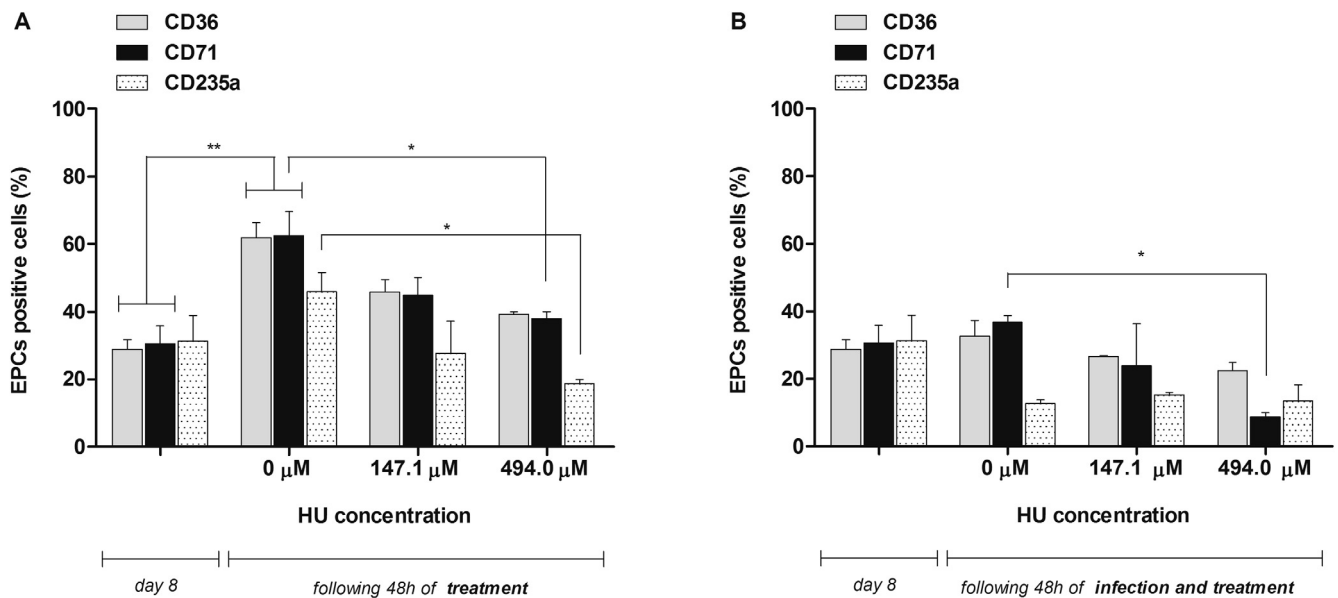


Fig. 6. Expression of the erythroid lineage specific markers CD36, CD71 and CD235a in EPCs by flow cytometry. Histograms are percentage values of positive cells, at day 8 from isolation, and after 2 days of treatment with HU at 494.0 μM and 147.1 μM (A), or following 2 days of infection and treatment with the different HU concentrations (B). Results are expressed as mean values $\pm$ standard error of means and are representative of two independent experiments. * $p < 0.05$, ** $p < 0.001$ indicates statistical differences between indicated dataset as determined using Dunnett's Multiple comparison test.

($p < 0.001$), while the remaining markers indicated a mere downward trend.

Finally, the distribution of EPCs in the various phases of the cell cycle after HU treatment and/or B19V infection was evaluated by PI staining and flow cytometry analysis. Table 1 reports the mean values from two independent experiments performed on EPCs cultures obtained from two different leukocyte-enriched buffy coats. The percentage rates of cells in G2/M increased in both

mock-infected and infected EPCs during the 48 h-period analyzed in the absence of HU [39]. By contrast, HU at the tested concentrations prevented cells from leaving the G1/S phases with a related reduction of cells in G2/M, in both experimental conditions. EPCs arrested with a 2 N DNA content may be not competent to engage in B19V active replication thus contributing to the antiviral effect of HU together with the lowering deoxyribonucleotide levels within cells.

Table 1

Cell cycle distribution of EPCs after HU treatment and B19V infection as determined by flow cytometry.

Cell population	EPCs (0 h)	Mock-infected EPCs (48 h)			B19V-infected EPCs (48 h)		
	0 μ M HU	0 μ M HU	147.1 μ M HU	494.0 μ M HU	0 μ M HU	147.1 μ M HU	494.0 μ M HU
G0/G1	77.9%	74.7%	70.5%	77.1%	73.0%	67.2%	76.7%
S	17.3%	17.0%	22.5%	19.2%	17.7%	24.3%	18.5%
G2/M	4.8%	7.6%	6.4%	3.5%	8.9%	7.2%	4.1%

4. Conclusions

The present study reports experimental evidence on the inhibitory effect of hydroxyurea towards B19V replication in the two relevant cellular model systems, UT7/EpoS1 and EPCs. HU displayed anti-B19V activity in absence of toxicity at concentrations measured in plasma samples of SCD patients undergoing HU therapy; this is of relevance in the perspective of its use in the treatment of B19V-caused diseases. In SCD population, the physiology of the HU effect is not completely understood but mainly relies in the enhanced production of fetal hemoglobin which helps prevent the formation of sickle-shaped red blood cells and improved blood flow through the circulatory system [35], thereby reducing the frequency of painful crises and the need for transfusions. Recent data indicate that HU therapy minimizes symptoms of worsened anemia during B19V-induced TAC in children with SCD [40]; patients who are treated with HU exhibit normal immunological response to a natural infection of B19V, and during a TAC episode experience attenuation of clinical symptoms with improved hematological indices and decrease frequency of RBCs transfusions when compared to HU-untreated children. HU can therefore show a dual beneficial effect on SCD patients, not only for the treatment of the disease but also towards a virus that is a relevant pathogen and cause of severe complications in the same patient population.

Conflict of interest

The authors declare no financial competing interests.

Acknowledgments

None.

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